

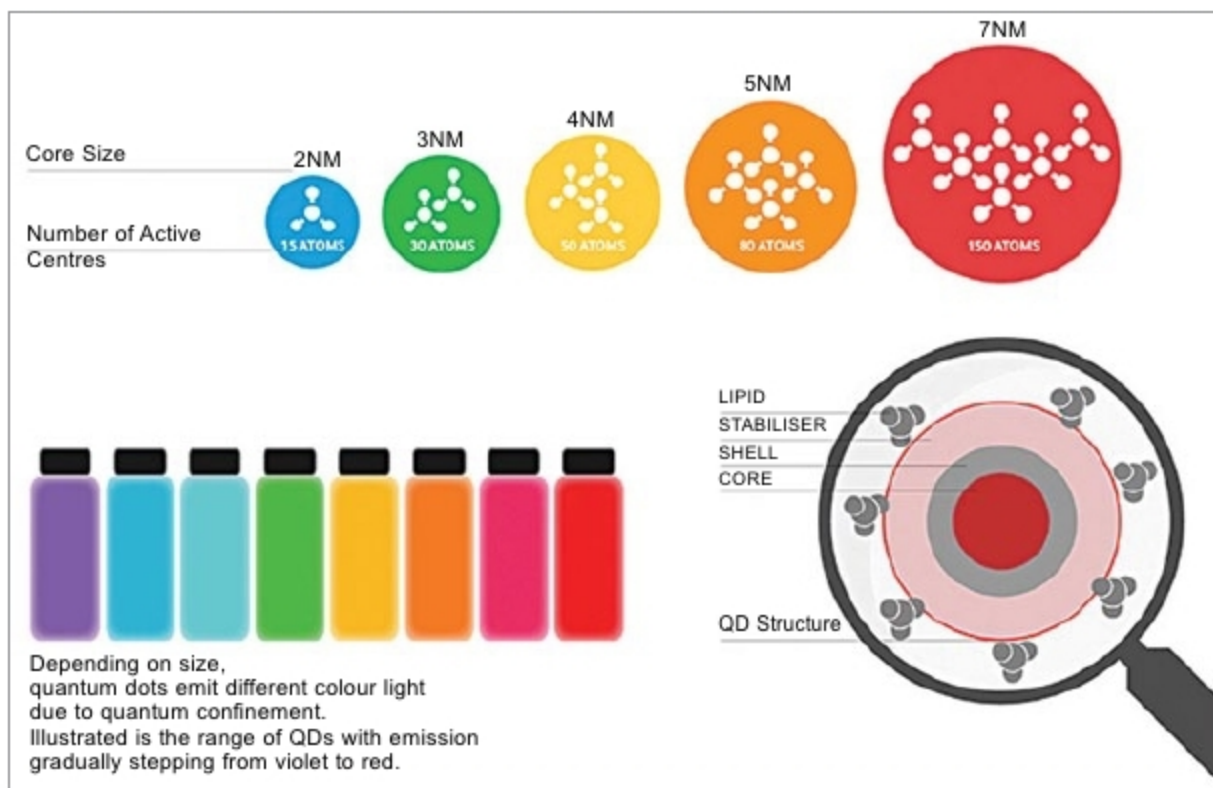


Fig. 3: Colour control by quantum dot particle size (Credit: <https://pid.samsungdisplay.com>)

LEDs, diode lasers, second-harmonic generation, quantum computing and medical imaging, among others.

- QDs generate two electrons for a single photon of light, as compared to normal solar cells that generate only one electron. Therefore use of QDs increases the efficiency of solar cells in producing electric power.
- QDs display consume less power than current displays. So these are a better choice for applications ranging from cellphones to large-screen TVs.

QDs versus other technologies

The use and application of QDs have many advantages. These require a small amount of energy for excitation regardless of size. This can be achieved by a single blue or ultraviolet wavelength beam. It reduces cost of sensing, too.

Being nanometre in size, QDs have a higher density of states than high-dimensional structures. This means electrons do not have to travel as far as with larger particles. Thus, electronic devices can operate faster.

S.K. Abdul Moyez, research scientist, University of Calcutta, says, "Application of QDs technology is growing widely from semiconducting engineering—including photovoltaic technology, display technology, mobiles and laptops, modern printing technology including ink-jet printing and 3D printing, electrical fields and ceramic engineering—to ultramodern

space research technology and cancer therapeutic technology due to their customisable band gap energy and high surface-to-volume ratio."

QDs are brighter and have little degradation over time, making these superior to traditional organic dyes used in biomedical applications. QD light has a sharp and narrow emission peak, due to a radiative recombination of an exciton. High peak brightness results in a large difference between the bright and dark parts of an image. This property makes QDs suitable for high dynamic range (HDR) displays.

Challenges with QD technology

Manufacturing of QDs is much more expensive than conventional LCDs. However, QDs are still cheaper than OLED displays. And with mass production, QDs display production will soon gain enough efficiencies to become relatively inexpensive.

QD particles are affected by water, heat and humidity, and require isolation. To protect QDs from decomposition due to humidity, a barrier film is currently used as an isolating layer.

Dr Mahesh Sharma, senior scientist, National Physics Laboratory, says, "The first challenge in QDs research is their mono-dispersivity. The other is stability. When kept for a long time, QDs can aggregate and form large clusters. To stabilise, ligands or any surface passivation must be used. This also sometimes interferes in certain

desired properties. Photocatalysis and water splitting are some challenging applications."

Dr Gupta adds, "Existence of free charge carriers and electric fields in LEDs presents a challenge for optimisation of QD band structure. Another big challenge for this technology is on-chip design, as it is hard to achieve. Manufacturing processes have reported challenges in the field of quantum displays, such as low quantum efficiency and need for an in-cell polariser."

Moyez adds, "Use of toxic but efficient materials in QDs is another factor to be considered for the sake of a green environment. QD polymer complex mixture is hard to develop. QDs in photovoltaic technology may create an excess recombination state, which could lower overall performance. The large size of QDs compared to biological cells is another factor that needs to be overcome."

Future of QD technology

The future of QD technology depends on the requirement of new applications and development areas. QDs are considered zero dimensional and, hence, exhibit sharper density of states in comparison with structures of higher dimensions. These result in excellent optical as well as transport properties. These properties are currently being studied for potential applications in electronic amplifiers, biological sensors and diode lasers.

Applications and usefulness of QDs technology continue to expand. Researchers are striving to bring the benefits of this technology to more technologically-applied fields. The broad range of real-time applications of QDs technology in the field of medical science is expected to be very useful in disciplines such as lymphocyte immunology, cancer metastasis, stem cell therapeutics and embryogenesis.

Development of QDs will change the face of upcoming technologies. Proper development or commercialisation of QDs in optical and semiconducting fields have already started due to the tunable band gap of QD materials. And more research in sectors such as science, engineering, technology and medicine is essential for a better future. **EFY**